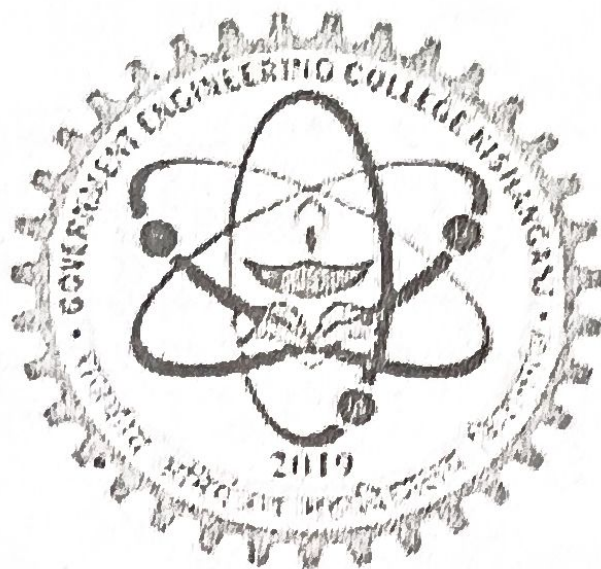


GOVT. ENGINEERING COLLEGE
KISHANGANJ, BIHAR



Assignment

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Branch: C.S.E Sem: 4th Year: 2026

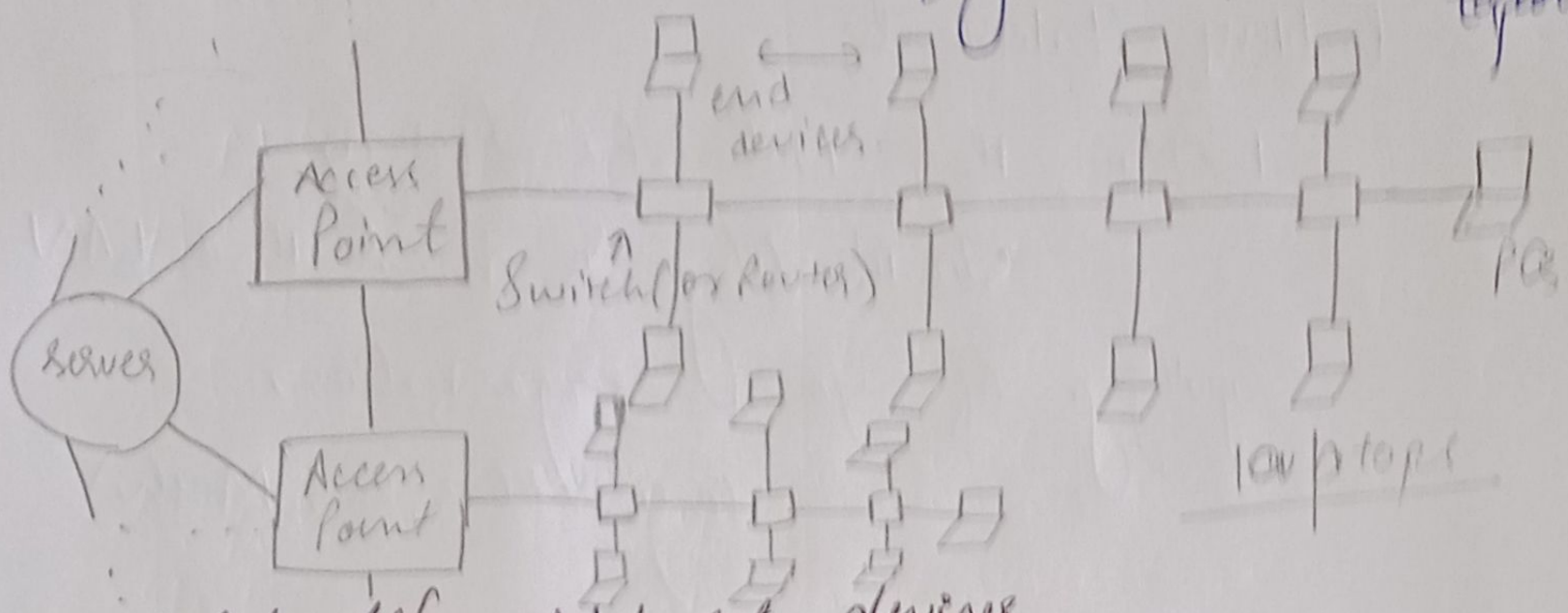
Session: 2024-28 Subject: Computer Network

Experiment Title: Study of LAN Topology, Network Devices and Address Configuration

Problem Statements

1. Draw the Network Topology
 - To identify & draw the LAN Topology of your organisation (e.g. star, bus, Mesh)
 - Labelling all components such as Router, Switch, Access Point and end devices (PCs, laptops).

- The LAN Topology of our organisation is like a bus topology. (It may be like a star, Mesh or hybrid topology)



Q. Identify Network devices

list all the networking devices used in the LAN

- Mention the function of each device.

Networking devices used are

- Switch, Router, Access point, end devices (PC, laptops, etc)

Functions of the above networking devices:

- Switch: A network switch connects multiple devices - like computers, printers and servers - within a single network.

- Router : A router is a device that connects two or more packet-switch networks or sub-networks.
- Access Point : It is a device that allows you to connect to a network (wired or wirelessly)
- End devices : laptops, pcs or smartboard are the end devices which uses the internet.

3. Analyze Connection Types:

- whether the connection is wired (ethernet) or wireless (wifi)

• Mention where each type is used

— the connection made by smartboard is of ethernet but the connection made by (desktops) pcs or laptops is wifi (wireless)

Roles of Switch, Router, Access Point & end devices

Switch

Roles

- Network Switch: It uses MAC addresses to filter and forward data packets strictly between the sender & the intended recipient, maximizing network speed & reducing collision.

- Router: It assigns local IP address to private addresses to public ones, and determines the most efficient path to route data packets to their final destination on the internet.

- Access Point: It receives, buffers and transmits the wired network infrastructure & wireless end devices (like smartphones or laptops) using radio waves.

End devices:

eg. computers, smartphones, smart TVs,
IoT devices & network printers

They run applications, generate data
requests and format information so it
can be sent across the network.

4. Measure Network Speed:

- Find the link speed of
 - wired or wireless network
- Mention the tools/commands used.

→ speed of wireless network is

for
Geek-Compuser
-wifi

Receive rate (Mbps): 57.8 or 433.3
Transmit rate (Mbps): 57.8 or 433.3

Command used is: netsh wlan show
interface

in windows
terminal

5. Find System Network Details

- Determine and record the following of your computer
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
 - Mention the command used
-

For Campus wife of GECK

IPv4 Address: 192.168.56.1

Subnet Mask: 255.255.255.0

MAC Address: 0C-4B-A1-29-36-CC

Command used: ipconfig, getmac -c

⑥ Difference between wired & wireless network

Feature	wired Network	wireless Network
Speed	Generally faster offering higher data transfer speeds.	often slower compared to wired, speed can fluctuate.
Reliability	More reliable, less prone to interference.	susceptible to interference and signal degradation.
Security	More secure It's physical network's access is required for unauthorised entry.	More vulnerable to unauthorised access and hacking.
Range	limited by the length of the cables.	limited by signal range affected by physical obstructions.
Bandwidth sharing	Each connection has its own bandwidth.	shared bandwidth.

• Role of Subnet-Mask in networking
Subnet Mask is a 32-bit number used with IP address to separate the network portion and host portion of an IP address

Other roles:-

• Identify network vs host

• Subnetting: Divides large network into smaller subnets to reduce traffic

• Routing Decision: Router uses subnet mask to check if destination is same network or not

• Finding broadcast Address

• Find no. of hosts.

```
PS C:\Users\harsh> netsh wlan show interfaces
```

There is 1 interface on the system:

Name	: Wi-Fi
Description	: Realtek RTL8851BE WiFi 6 802.11ax PCIe Adapter
GUID	: d27472da-2071-466a-8132-6eb48072a4b6
Physical address	: dc:4b:a1:29:36:cc
Interface type	: Primary
State	: connected
SSID	: GECK-Campus-WiFi
AP BSSID	: b0:19:21:d4:91:9f
Band	: 5 GHz
Channel	: 161
Connected Akm-cipher	: [akm = 00-0f-ac:02, cipher = 00-0f-ac:04]
Network type	: Infrastructure
Radio type	: 802.11ac
Authentication	: WPA2-Personal
Cipher	: CCMP
Connection mode	: Auto Connect
Receive rate (Mbps)	: 433.3
Transmit rate (Mbps)	: 433.3
Signal	: 100%
Rssi	: -56
Profile	: GECK-Campus-WiFi
QoS MSCS Configured	: 0
QoS Map Configured	: 0
QoS Map Allowed by Policy	: 0

PS C:\Users\harsh> getmac

Physical Address	Transport Name
=====	=====
DC-4B-A1-29-36-CC	\Device\Tcpip_{D27472DA-2071-466A-8132-6EB48072A4B6}
0A-00-27-00-00-11	\Device\Tcpip_{7B3F6111-677B-4614-9B7A-ADF21DDA9C3D}
PS C:\Users\harsh>	

PS C:\Users\harsh> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::b0fc:673:3301:1b7%17
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Wi-Fi 3:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 5:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : mshome.net
Link-local IPv6 Address : fe80::87d1:21f7:6a75:bbe3%21
IPv4 Address. : 10.85.19.40
Subnet Mask : 255.255.248.0
Default Gateway : 10.85.16.1

PS C:\Users\harsh>